



How can India strengthen fire safety systems to protect highly vulnerable patients in ICUs?

Electrical Overload • RACE Protocol • Electrical Harmonics • Layered Protection • Power Audits

GS II
Health
GS III • Disaster
Management

📖 KNOWLEDGE POINTS

- **ICU Fire Vulnerability:** ICUs combine dense electrical equipment, continuous power demand, oxygen-enriched conditions and patients unable to self-evacuate. Enclosed layouts can rapidly accumulate smoke, making ordinary building-level fire precautions insufficient for protecting critically ill patients.
- **Electrical Overloading Risk:** Modern hospitals depend on ventilators, warmers, monitors and other high-load devices. Increased appliance density, ageing wiring, overloaded circuits and inadequate maintenance can generate excessive heat, insulation failure or electrical faults capable of initiating fires.
- **RACE Protocol:** The RACE framework structures immediate hospital fire response: Rescue people in danger, raise the Alarm, Confine fire and smoke, and Extinguish the fire when safe or Evacuate occupants according to established procedures.
- **Electrical Harmonics:** Electronic medical equipment can draw distorted electrical currents, creating harmonics that increase heating in neutral conductors, transformers and connections. Such degradation may develop without conventional breakers tripping, making specialised power-quality audits important for prevention.
- **Layered Fire Protection:** Effective ICU safety requires multiple independent safeguards rather than reliance on one measure. Safe electrical design, alarms, sprinklers, compartmentation, accessible exits, evacuation protocols, equipment maintenance and trained personnel collectively reduce both fire probability and consequences.

📄 QUICK FACTS

- In August 2026, a fire in the NICU of Amravati District Women's Hospital killed three newborns; a ventilator-related fault was suspected.

The 2026 National Guidelines on Fire and Life Safety in

- Healthcare Facilities specifically cover high-risk areas including ICUs, NICUs, PICUs and operation theatres.

Following the August 2026 Amravati NICU tragedy, Nashik

- authorities directed government and private hospitals to conduct immediate fire and electrical safety audits of ICUs.

The National Neonatology Forum's fire-safety advisory

- recommends mandatory fire-safety certification for hospitals, especially those with NICUs, alongside periodic mock drills and inspections.

🎯 KEY TAKEAWAY

India's healthcare fire safety challenge is fundamentally about safeguarding the most vulnerable through preventive governance, technological reliability and institutional preparedness.